PROJECT DOCUMENT

BIOMASS ENERGY & AI

Reforesting in Mitsue

Cashflow Model — Inflow vs. Outflow

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BIOMASS ENERGY & AI — Cashflow Model

Reforesting in Mitsue · Inflow planned against the monthly cost baseline

*This document adds the **cash-in** side to the project's already-detailed **cash-out** timeline (the EVM monthly S-curve, `mitsue_evm_plan.md` §4). Its single design rule: **the cumulative balance must never fall to zero in any month.** Outflow is smooth and monthly; inflow arrives in **big, lumpy chunks** — founder capital, grants, subsidies, corporate/loan tranches — until operating revenue begins near the end of Phase 3.*

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1. The design rule

> **Every month: cumulative cash-in − cumulative cash-out ≥ a positive working-capital floor.**

Because spending is continuous but funding arrives in lumps, each tranche must **land in or before the month the balance would otherwise breach the floor.** We target a **minimum floor of ~¥3–5M** early (Phases 0–1, when monthly burn is tiny) and a larger absolute buffer through Phase 3 (when burn peaks at ¥25M/month — a single late tranche can blow a ¥25M hole in one month).

All figures are illustrative, consistent with the EVM PMB (BAC ¥220M) and the funding stack (¥192M target, ¥28M gap if the target is fully realized). **None of the ¥192M is secured yet** — including the ¥6M founder-capital layer. They set the **shape and timing** of inflow, not vetted amounts — Phase 1 feasibility + confirmed grant awards replace them at Baseline Rev 2 (Dec 2026).

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2. Cash-out recap (from EVM §4)

Monthly planned spend, front-loaded on studies/design, peaking in construction (M22–M28). Cumulative reaches the ¥220M BAC at M30.

Phase	Months	Spend in phase	Cum. spend by phase-end
P0 Pre-Foundation	M1–M3	¥0.25M	¥0.25M
P1 Foundation	M4–M9	¥5.5M	¥5.75M
P2 Pilot Design	M10–M18	¥22.5M	¥28.25M
P3 Pilot Build	M19–M30	¥191.75M	¥220.0M

Peak monthly burn: **¥25M/month at M25–M26** (Apr–May 2028).

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3. Inflow tranche schedule (recommended)

Funding is drawn down as **11 lumps**, each timed to a funding gate or to the month a spend wave begins. Sources map to the funding stack (L1–L5) plus a bridge/working-capital tranche.

Month	Cal	Tranche	¥M	Source layer	Rationale / trigger

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M1	Apr 2026	Seed	6	L1 Founders	Founder capital — sole runway for Phases 0–1
M8	Nov 2026	Planning grant	5	L2 Gov	First small grant, unlocked by early feasibility work; lands before Gate 2
M9	Dec 2026	Foundation 1	15	L3 Foundations	Legal entity registered (P1) unlocks foundation grants
M12	Mar 2027	Gov grant 1	20	L2 Gov	Feasibility studies unlock national/pref. grants
M15	Jun 2027	Corporate 1	15	L4 Corporate	CSR / prepaid hosting / minority GK equity
M18	Sep 2027	交付金 / Gov 2	30	L2 Gov	MoE decarbonization 交付金, first disbursement via village
M21	Dec 2027	交付金 / Gov 3	40	L2 Gov	Main capex subsidy — **lands at Phase 3 start**, before peak burn
M23	Feb 2028	Corporate 2	22	L4 Corporate	Equity/offtake tranche 2 (post-pilot-progress)
M25	Apr 2028	交付金 / Gov 4	30	L2 Gov	Capex subsidy tranche 2 at peak-burn month
M26	May 2028	**Bridge / gap**	18	Loan / MR	**Working-capital + gap-closing** (see §5)
M27	Jun 2028	Foundation 2	25	L3 Foundations	Second foundation tranche / gap-closing
M28–M30	Jul–Sep 2028	Operating revenue	3	L5 Revenue	Pilot online — first energy/DC/EV income (¥1M/mo)

Month	Cal	Tranche	¥M	Source layer	Rationale / trigger
		Total in	**229**		Covers ¥220M BAC + ~¥9M end buffer

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4. Monthly running balance (recommended schedule)

The balance stays **positive every month**, with the thinnest point at **M7 (¥3.05M)** — just before the first grant lands.

Month	Cal	Cash-out (mo)	Cum-out	Cash-in (mo)	Cum-in	Balance
M1	Apr'26	0.05	0.05	6	6	**5.95**
M2	May'26	0.10	0.15	—	6	5.85
M3	Jun'26	0.10	0.25	—	6	5.75
M4	Jul'26	0.20	0.45	—	6	5.55
M5	Aug'26	0.50	0.95	—	6	5.05
M6	Sep'26	0.80	1.75	—	6	4.25
M7	Oct'26	1.20	2.95	—	6	**3.05 ◀ floor**
M8	Nov'26	1.50	4.45	5	11	6.55
M9	Dec'26	1.30	5.75	15	26	20.25
M10	Jan'27	1.50	7.25	—	26	18.75
M11	Feb'27	2.20	9.45	—	26	16.55
M12	Mar'27	2.80	12.25	20	46	33.75
M13	Apr'27	3.00	15.25	—	46	30.75
M14	May'27	3.00	18.25	—	46	27.75
M15	Jun'27	3.00	21.25	15	61	39.75
M16	Jul'27	2.50	23.75	—	61	37.25
M17	Aug'27	2.50	26.25	—	61	34.75
M18	Sep'27	2.00	28.25	30	91	62.75
M19	Oct'27	4.00	32.25	—	91	58.75
M20	Nov'27	7.00	39.25	—	91	51.75
M21	Dec'27	11.00	50.25	40	131	80.75
M22	Jan'28	16.50	66.75	—	131	64.25
M23	Feb'28	20.50	87.25	22	153	65.75
M24	Mar'28	20.50	107.75	—	153	45.25
M25	Apr'28	25.00	132.75	30	183	50.25
M26	May'28	25.00	157.75	18	201	43.25
M27	Jun'28	20.50	178.25	25	226	47.75
M28	Jul'28	20.50	198.75	1	227	28.25

Month	Cal	Cash-out (mo)	Cum-out	Cash-in (mo)	Cum-in	Balance
M29	Aug'28	14.00	212.75	1	228	15.25
M30	Sep'28	7.25	220.00	1	229	**9.00**

Two binding constraints the schedule is built around:

1. **M7 (Oct 2026) — the early pinch.** Planned founder capital (¥6M, not yet committed) is modeled as the *only* money in for the first 7 months. If it is not actually secured by then, or the M8 planning grant slips, or early spend overruns, the balance breaches zero here. *Mitigant: commit founder/seed capital ≥¥6M before Phase 0 spend begins, or secure a small Gate-1 bridge by M7.*
2. **M21–M27 (Phase 3) — the burn cliff.** At ¥20–25M/month, cash-in timing risk is severe: a one-month grant delay = a ¥25M hole. Every Phase-3 tranche must be **contracted well ahead of its landing month**, because grants often pay *in arrears*. *Mitigant: front-load the 交付金 tranche to M21 (Phase-3 start) and hold the bridge (§5) as the shock absorber.*

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5. The target-stack-only reality: where it runs dry

This is the **best case**, not a floor: it assumes the full ¥192M target stack lands exactly as planned, with no gap-closing tranche. **None of the ¥192M is secured today.** Even in that best case, the balance goes **negative at M28 (Jul 2028)** and bottoms at **−¥28M at M30** — i.e. the end-of-project shortfall is *exactly* the known ¥28M funding gap. If the target stack itself is not fully realized, the shortfall is larger and arrives sooner.

Month	Cum-out	Cum-in (target stack only)	Balance
M26	157.75	171	+13.25
M27	178.25	189	+10.75
M28	**198.75**	**190**	**−8.75 ◀ runs dry**
M29	212.75	191	−21.75
M30	220.00	192	**−28.00** (= the gap)

Implication: the ¥28M gap is not an abstract "shortfall vs BAC" — it is a **cash wall in July 2028.** To keep the balance positive through construction, the gap must be **secured *and* scheduled to disburse before M28**, and prudently by **Gate 3 (M21)** given the pay-in-arrears risk. This is why §3 places a bridge/gap tranche at M26 and a foundation tranche at M27.

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6. Initial inflow = chunks, not monthly

As the framing notes, early inflow is **lumpy, not monthly**:

- **Phases 0–2 (M1–M18):** 100% of inflow is founder capital + grants + subsidies + corporate/loan tranches. **Zero operating revenue.** Six lumps totalling ¥91M carry the project to the start of construction.
- **Phase 3 (M19–M30):** still dominated by lumps (¥135M across four tranches), with the **first trickle of operating revenue (¥1M/month) only from M28** as the pilot energises.
- **Phase 4 (M31+, out of this baseline):** the model flips — operating revenue (DC hosting, heat, EV, broadleaf/J-Credit, plus **grid-export electricity — likely FIP**, the plant being an exporter >1,000 kW) becomes the primary monthly inflow and funds the CHP scale-up. The co-located anchor DC is only a small behind-the-meter baseload (~2% of the CHP's own output); most of the *plant's* generation is village + export, so grid

capacity is a real dependency. That 2% is not a ceiling on DC hosting revenue, though — the larger showcase DC (old school) runs on grid/green power, and compute earns ~6–20× more per kWh than FIP either way, so hosting fill-rate — not CHP export share — drives the DC hosting line. See `mitsue_revenue_model.md` and EVM §14.

Loans/bridge financing matter precisely at the seams between lumps. Because grants commonly pay *on milestone completion* (arrears), a **bridge facility** sized to ~1 month of peak burn (**~¥25M**) lets the project spend against a *committed-but-not-yet-disbursed* grant without the balance dipping. The M26 tranche in §3 plays this role; it can be a bank bridge loan, a draw on the ¥25M Management Reserve, or an advance against a signed 交付金.

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7. Recommendations

1. **Adopt the ~¥3–5M early floor and the Phase-3 buffer as hard rules** in the monthly EVM report — track *balance*, not just spend.
2. **Close the ¥28M gap before Gate 3 (M21),** not at end-of-project. Until the target stack is actually secured, the target-stack-only line (§5) is the *best-case* forecast and it still fails in M28.
3. **Contract every Phase-3 tranche 2–3 months ahead of its landing month** to absorb grant pay-in-arrears lag.
4. **Arrange a ~¥25M bridge facility** (bank line, MR draw, or grant advance) as the shock absorber for the burn cliff.
5. **Re-time this model to the live Gantt at Baseline Rev 2 (Dec 2026)** alongside the cost re-sync, and fold in confirmed 交付金 amounts and any loan terms.

> **Status — adopted.** All five are now encoded as governing rules in the EVM plan (v2.7): liquidity floor → §8.1; monthly balance-vs-floor reporting → §9; gap-before-Gate-3, tranche 2–3-month lead time, and ~¥25M bridge facility → §10 assumptions/constraints; cashflow re-time + 交付金/loan terms → §12 Rev 2 scope. The Gate-3 timing rule is also flagged in the funding flowchart (v2.8). *Remaining as a live action:* actually arranging the bridge facility (bank line / MR / grant advance) before Gate 3 — a Phase 2 funding task, not yet secured.

Sources & status

Cash-out figures: `mitsue_evm_plan.md` §4 (Baseline Rev 1). Funding stack: `mitsue_phases_funding_flowchart.md` (¥192M target, ¥0 secured, ¥28M gap if fully realized). Tranche amounts and months are illustrative planning figures, not committed disbursement dates — to be replaced by confirmed grant/loan schedules at Baseline Rev 2. Operating-revenue onset per `mitsue_revenue_model.md`.

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